DAY 6 OF 30

# Delta tables

The format under every table.
Why it matters more than it looks.

**Parquet** **Transaction log** **Time travel**

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THE PROBLEM

# Plain files in a lake have no safety net.

| | | |
|---|---|---|
| 1 | **Job fails midway** | Half-written data stays |
| 2 | **Two writers at once** | They overwrite each other |
| 3 | **A bad load** | No way to go back |
| 4 | **A changed column** | Readers quietly break |

Files are cheap and open, but **nobody is keeping the books**.

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THE SIMPLE VERSION

# A notebook that keeps every edit.

Imagine a ledger where nothing is erased. Each change is a new dated entry, so you can always see what the page said last week.

A Delta table is **data files plus that ledger**.

| | |
|---|---|
| ✓ **Data files** | Stored as Parquet |
| ✓ **The ledger** | A transaction log |
| ✓ **Together** | A table you can trust |

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HOW IT WORKS

# What a write really does

1. New Parquet files are written **next to** the old ones
2. A new entry is added to the **_delta_log** folder
3. The entry lists which files are **in** and which are **out**
4. Readers follow the log, so they see **all or nothing**

**Why it matters**

A failed job leaves no entry in the log, so readers never see the half-written files.

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FOR ENGINEERS

# What you get from the log

### ACID transactions

Concurrent readers and writers see a consistent table, even while a load is running.

### Time travel

Query the table as of an earlier version or timestamp. Useful for audits and for undoing mistakes.

### Schema enforcement

A write with the wrong columns is rejected instead of silently corrupting the table.

### Open format

The same files are read by Spark, the SQL engines and Power BI in Fabric.

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THE HONEST PART

# What Delta will not do for you

| IT HANDLES | YOU STILL OWN |
| --- | --- |
| Consistency during writes | Compacting many small files |
| History and rollback | Cleaning up old versions |
| Schema checks on write | Choosing a sensible layout |

> **The catch:** every write adds files. A table loaded every few minutes can collect thousands of tiny files and get slower each day until it is compacted.

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FROM REAL WORK

# Delta on a real migration

A warehouse migration to Databricks for an energy company in Europe.

1. **SQL stored procedures** were rewritten in PySpark
2. **Delta Lake** gave the new pipelines ACID reliability
3. **Processing time** went from 2 hr 50 min to about 5 min
4. **Rebuilt data models** fed KPIs into Power BI

Different platform, **same format** Fabric is built on.

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SAVE THIS

# Delta in five lines

1. **Parquet files** hold the data
2. **A transaction log** records every change
3. **Readers** see complete versions only
4. **Time travel** lets you look back
5. **Maintenance** is still your job

## Tomorrow: Shortcuts.
## Using data that lives elsewhere, without copying it.

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Follow for Day 7